

Vaibhav Saini

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Summary

Software Engineer (Backend & AI) with experience designing and building scalable backend systems and production-grade APIs using Python and FastAPI. Strong in microservices architecture, AI/ML model integration, and cloud-based deployments. Focused on writing clean, maintainable code and improving system performance, reliability, and scalability.

Education

- Master's in Information Technology (Security) May 2026 | GPA: 3.96
Arizona State University, Arizona, US
- Bachelor's in Computer Science (Data Science) June 2023 | GPA: 7.62
MMDU, Ambala, Haryana, India

Certifications

- AWS Certified Cloud Practitioner - Amazon Web Services (AWS) July, 2025
- AWS Academy Graduate - Cloud Architecting - Training Badge April, 2025
- Machine Learning Specialization - DeepLearning.AI January, 2026
- Generative AI with LLMs - DeepLearning.AI February, 2026

Technical Skills

- Languages & Frameworks: TypeScript, React, Node.js, Python, FastAPI, .NET
- AI & ML: LLMs, Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, Model Integration
- Full-Stack & APIs: REST APIs, Backend Architecture, Frontend Development, Real-Time Systems
- Cloud & DevOps: AWS, Azure, and Docker, CI/CD (basic), Cloud Deployment
- Database Management: MySQL, MongoDB, DynamoDB, and PostgreSQL
- Security & Monitoring: Network Security, Threat Detection, Logging, Observability, Wireshark, and Splunk
- Other: System Design, Observability, Logging, Performance Optimization
- **Systems:** Distributed Systems Basics, API Design, Performance Optimization

Selected Projects

SecRAG: Retrieval-Augmented Generation (RAG) Backend | FastAPI, Sentence-Transformers, NumPy

- Designed and built a scalable backend service for document-based question answering using FastAPI
- Developed RESTful APIs for document ingestion, query processing, and response generation in a microservices architecture
- Integrated AI/ML models (Sentence-Transformers embeddings) into backend pipelines for semantic retrieval
- Optimized retrieval performance using hybrid search (BM25 + vector similarity)
- Implemented structured metadata tracking to improve reliability, traceability, and system performance

PulseScore: Real-Time AI Inference & Monitoring System | FastAPI, Python, ML Deployment

- Built a real-time AI inference backend system with low-latency prediction APIs
- Integrated machine learning models into production-style backend services
- Designed logging and monitoring pipelines to improve system reliability and debugging
- Optimized API performance for continuous high-frequency requests
- Implemented versioned model deployment pipeline for maintainability and scalability

Secure Investment Platform | Angular, .NET 8, Docker, Azure

- Built a full-stack fintech application supporting 100+ users with high request throughput
- Implemented secure authentication and role-based access control (RBAC)
- Deployed containerized services using Docker on Azure with 99.9% uptime
- Improved system reliability and observability through monitoring and logging tools

Relevant Experience

Independent Backend & AI Development

- Designed and deployed multiple backend systems using FastAPI and Python
- Built and tested REST APIs for real-time data processing and AI inference
- Worked on end-to-end system design including data handling, model integration, and monitoring
- Focused on building production-style systems rather than academic prototypes